

Notes: Calvet, Campbell, and Sodini (JPE, 2007)

Down or Out: Assessing the Welfare Costs of Household Investment Mistakes

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1 Big picture

- **Question.** How costly are household portfolio mistakes once we measure them using detailed security-level data rather than broad survey categories? The paper studies two mistakes: *underdiversification* (“down”) and *nonparticipation* in risky asset markets (“out”).
- **Setting.** The paper uses Swedish administrative records for essentially the whole population, with individual-level holdings of bank accounts, mutual funds, and stocks outside defined-contribution pension accounts. The key period is 1999–2002, with most portfolio calculations focused on end-2002 positions.
- **Core empirical challenge.** Most existing data either do not observe complete portfolios, or do not identify the underlying securities. Without complete security-level data, it is hard to know whether households are truly badly diversified or whether they just hold diversified mutual funds and cash.
- **Main conceptual contribution.** The paper insists that diversification has to be measured at the level of the *whole risky portfolio*, not just directly held stocks. This is why observing mutual funds matters so much.
- **Main result on underdiversification.** A minority of households are very poorly diversified, but the *median* Swedish household is not. Most participating households hold enough cash and internationally diversified mutual funds that their total losses from underdiversification are modest.
- **Main result on nonparticipation.** The standard welfare calculation for staying out of the stock market assumes that nonparticipants would invest efficiently if they entered. The paper argues this is too optimistic. Once we recognize that nonparticipants would likely invest inefficiently as well, the cost of nonparticipation falls by almost one-half.
- **Headline quantitative results.**
 - Median risky-portfolio volatility is about 19.5% per year.
 - More than half of the risk in the median risky portfolio is idiosyncratic.
 - Median complete-portfolio return loss from underdiversification is about 1.17% of financial wealth, roughly \$131 per year, relative to the currency-hedged world index.
 - The standard estimate of the cost of nonparticipation is about 4.3%, but falls to about 2.3% when the likely inefficiency of nonparticipants is taken into account.
- **Why this matters.** The paper weakens the common claim that “households are obviously making gigantic diversification mistakes.” The right statement is more nuanced: many households invest reasonably well, but a nontrivial minority makes very costly mistakes.
- **Economic takeaway.** Unsophisticated households tend to be both less diversified and less aggressive. Richer and more educated households invest more efficiently, but they also take more risk. So sophistication improves the *quality* of the risky portfolio, but not necessarily the total return loss on the complete portfolio.

2 Data and measurement (key objects)

2.1 Administrative household panel

- **Source.** Swedish government wealth-tax records, combined with income and demographic records.
- **Coverage.** The panel covers all Swedish households from 1999 to 2002. Because Sweden levied a wealth tax, the government collected detailed records on financial assets held outside defined-contribution pension accounts.
- **What is observed.**
 - bank accounts,
 - money market funds,
 - mutual funds,
 - directly held stocks,
 - capital insurance products (total value but not internal composition),
 - debt,
 - real estate,
 - disposable income,
 - age, education, immigration status, household size, and related demographics.
- **Security-level detail.** Financial institutions report holdings down to the individual security or account level, which is the crucial advantage of the data.
- **What is missing.** Defined-contribution pension wealth is not directly observed. The paper notes that this missing category was about \$25.6 billion in 2002, while observed nonretirement financial wealth was about \$131.3 billion, so the data still cover about 84% of household financial wealth outside retirement saving. Since pension saving is usually in mutual funds, including it would likely make households look *more* diversified, not less.

2.2 Aggregate household balance sheet in 2002

- Aggregate gross household wealth in the micro data is about \$478.7 billion.
- Financial wealth is about \$131.7 billion, or 27.5% of gross wealth.
- Real estate is about 72.5% of gross wealth.

Within financial wealth:

- cash (bank accounts plus money market funds) is about 41%,
- mutual funds are about 22%,
- direct stockholdings are about 23%,
- capital insurance is about 9%,
- bonds and derivatives are about 5%.

Interpretation. The aggregate household portfolio is not mostly direct stockholding. This is why a “direct-stocks-only” view exaggerates underdiversification.

2.3 Cross-sectional portfolio composition

Figure 1 in the paper shows how the composition of financial wealth changes across the gross-wealth distribution:

- for most households, mutual funds are the largest risky asset category,
- in the very top of the wealth distribution, direct stockholdings become dominant,
- the richest households end up holding more than half of financial wealth in direct stocks,
- but mutual funds dominate stocks in most household portfolios.

Interpretation. The paper is not saying “everyone is diversified.” It is saying that most *middle-class* households obtain diversification through funds, whereas the very rich hold much more concentrated direct stock portfolios.

2.4 Micro sample used for portfolio analysis

- The paper’s core cross-sectional analysis uses a random sample of 100,000 households at the end of 2002.
- Households with extremely low income or financial wealth and those with hard-to-measure short-lived assets are excluded in the main analysis.

Important summary statistics:

- About 62% of households participate in risky asset markets.
- Median complete financial portfolio: about \$12,338.
- Median risky portfolio among participants: about \$4,295.
- Participants are much wealthier than nonparticipants on average.

Interpretation. Participation is common but far from universal, and the typical risky portfolio is not huge in dollar terms.

2.5 Key measurement choices

The paper studies diversification within portfolios of:

- stocks,
- mutual funds (both bond and equity funds),
- and cash.

It excludes from the main diversification calculations:

- human capital,

- real estate,
- bonds and derivatives held directly,
- and the inside composition of capital insurance products.

Interpretation. This is a financial-portfolio analysis, not a full household balance-sheet portfolio problem. The authors openly say this is a limitation, but also argue that the financial-portfolio question is already very informative.

2.6 Why mutual funds are the key object

The paper emphasizes that popular Swedish mutual funds were internationally diversified. For example, popular funds sold by large domestic banks held substantial foreign equity shares. This makes it relatively easy for middle-class households to diversify internationally through funds.

Implication. Direct stock concentration alone is an unreliable measure of household inefficiency. Once mutual funds are observed, the picture is much less pessimistic.

3 Models and empirical specifications (all equations + interpretation)

3.1 Portfolio return decomposition

Let q_h be the vector of portfolio weights for household h , and let Σ be the variance-covariance matrix of asset excess returns. Then household portfolio variance is

$$\sigma_h^2 = q_h' \Sigma q_h. \quad (1)$$

Interpretation. Since the data observe holdings but not realized within-year rebalancing, the paper infers the risk of each household portfolio from the covariance matrix of the underlying assets.

3.2 Systematic and idiosyncratic risk

Given a benchmark portfolio B , the paper runs

$$r_{h,t}^e = \alpha_h + \beta_h r_{B,t}^e + \varepsilon_{h,t}, \quad (1)$$

where $r_{h,t}^e$ is the household portfolio excess return and $r_{B,t}^e$ is the benchmark excess return.

This implies the variance decomposition

$$\sigma_h^2 = \beta_h^2 \sigma_B^2 + \sigma_{i,h}^2, \quad (2)$$

where $\sigma_{i,h}$ is the idiosyncratic volatility of household h 's risky portfolio.

The corresponding idiosyncratic variance share is

$$\frac{\sigma_{i,h}^2}{\sigma_h^2}. \quad (3)$$

Interpretation.

- Systematic risk is the part correlated with the global equity benchmark.
- Idiosyncratic risk is unrewarded in the CAPM benchmark the paper adopts.
- Therefore, idiosyncratic risk is interpreted as an investment mistake.

3.3 Idiosyncratic risk decomposition inside the risky portfolio

To understand why some households have more idiosyncratic risk than others, the paper decomposes the risky portfolio into:

- average idiosyncratic volatility of assets in the portfolio,
- concentration of holdings,
- and average correlation among the idiosyncratic components of the assets.

Under a stylized symmetric approximation,

$$\sigma_{i,h}^2 = \sigma_{a,h}^2 [C_{a,h} + (1 - C_{a,h})\rho_{a,h}], \quad (4)$$

where:

- $\sigma_{a,h}^2$ is average asset-level idiosyncratic variance,
- $C_{a,h}$ is a concentration index,
- $\rho_{a,h}$ is the average correlation among idiosyncratic components.

A log-linearization yields equation (5) in the paper.

Interpretation. High idiosyncratic risk can come from holding volatile assets, holding too few assets, or holding assets that move together. Empirically, the paper finds that average asset volatility is the dominant component.

3.4 Benchmark asset-pricing model for mean returns

Because realized histories for many assets are short, the paper does not estimate mean returns directly from sample averages. Instead it imposes the global CAPM in dollars:

$$r_{j,t}^e = \beta_j r_{m,t}^e + \varepsilon_{j,t}, \quad (6)$$

where $r_{m,t}^e$ is the excess return on the world market portfolio relative to the U.S. Treasury bill.

The benchmark world index is the MSCI All Country World Index. From the point of view of a Swedish investor, the paper considers two benchmark versions:

- the **unhedged world index**, which carries currency risk,
- the **currency-hedged world index**, which under covered interest parity is mean-variance efficient in the model.

For 1983–2004, the paper reports for the MSCI world index in U.S. dollars:

- mean excess return = 6.7%,

- standard deviation = 14.7%,
- Sharpe ratio = 45.2%.

Interpretation. The whole welfare analysis depends on a benchmark frontier. The hedged world index is the efficient benchmark under the maintained model; the unhedged world index is a more realistic benchmark for households who do not hedge currency exposure.

3.5 Diversification losses: Sharpe-ratio loss

The first measure of inefficiency is the relative Sharpe ratio loss:

$$RSRL_h = 1 - \frac{S_h}{S_B}, \quad (7)$$

where S_h is the Sharpe ratio of the household portfolio and S_B is the Sharpe ratio of the benchmark. If the benchmark is mean-variance efficient,

$$(1 - RSRL_h)^2 = 1 - \frac{\sigma_{i,h}^2}{\sigma_h^2}. \quad (8)$$

The paper also shows that in this case

$$RSRL_h = 1 - \text{Corr}(r_{h,t}^e, r_{m,t}^e). \quad (9)$$

Interpretation.

- Relative Sharpe-ratio loss is a pure efficiency measure.
- It is independent of the assumed equity premium.
- When the benchmark is efficient, it maps directly into the household's idiosyncratic risk share.

3.6 Diversification losses: return loss

The second measure is the return loss at the same risk level:

$$RL_h = w_h(S_B\sigma_h - \mu_h), \quad (10)$$

where w_h is the share of risky assets in the complete portfolio.

The paper gives two decompositions:

$$RL_h = S_B w_h \sigma_h RSRL_h, \quad (11)$$

and

$$RL_h = \mathbb{E}[r_m^e] w_h \beta_h \left(\frac{RSRL_h}{1 - RSRL_h} \right). \quad (12)$$

Interpretation.

- Sharpe-ratio loss measures portfolio inefficiency only.
- Return loss multiplies that inefficiency by how aggressively the household invests.
- Therefore, conservative but inefficient households can have small return losses, while sophisticated but aggressive households can have larger return losses even if their risky portfolios are better diversified.

3.7 Decomposition of complete return loss

Taking logs of equation (12), the paper obtains

$$\ln RL_{complete,h} = \ln \mathbb{E}[r_m^e] + \ln w_h + \ln \beta_h + \ln \left(\frac{RSRL_h}{1 - RSRL_h} \right). \quad (13)$$

This decomposition separates:

- **aggressiveness** — w_h and β_h ,
- **portfolio inefficiency** — the nonlinear function of $RSRL_h$.

Interpretation. This is the central accounting identity behind Table 5 and Figure 3. It explains why more sophisticated households can have lower Sharpe-ratio loss but higher total return loss.

3.8 Participation and risky-share regressions

The paper studies participation with a probit model and studies risky share and Sharpe ratio among participants with OLS regressions on household characteristics.

The right-hand-side variables include:

- disposable income,
- financial wealth,
- real estate wealth,
- liabilities,
- private pension contributions as a share of income,
- retirement, unemployment, entrepreneurship, and student dummies,
- age,
- household size,
- education,
- immigration.

Interpretation. These regressions do not “identify” causal effects in the strict modern reduced-form sense. Rather, they recover who is more likely to participate, who takes more risk, and who appears more diversified.

3.9 Welfare cost of nonparticipation

The return cost of nonparticipation for household h is defined as

$$RC_{complete,h} = S_h^* w_h^* \sigma_h^*, \quad (14)$$

where S_h^* and $w_h^* \sigma_h^*$ are the hypothetical Sharpe ratio and volatility the household would choose if it entered risky asset markets.

The paper studies two scenarios:

1. **Efficient-entry scenario:** nonparticipants would invest like an efficient benchmark investor.
2. **Realistic-entry scenario:** nonparticipants would invest like observably similar participants, using fitted values from the OLS regressions for risky share and Sharpe ratio.

Interpretation. The gap between these two scenarios is the entire point of the “down or out” title: households that are currently out might do badly if they came in.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the diversification results

The diversification analysis is credible because it combines three unusual advantages:

- complete population-level administrative coverage,
- security-level portfolio holdings,
- and direct observation of mutual fund positions and cash holdings.

This avoids the main weaknesses of earlier data:

- surveys do not observe exact securities,
- brokerage accounts do not observe total portfolios,
- ownership registries typically miss mutual funds or other assets.

Why this matters. The paper’s main empirical message – that most households are not disastrously underdiversified once mutual funds and cash are counted – depends exactly on these data advantages.

4.2 What identifies mean returns

The paper does *not* estimate expected returns from short individual-asset samples. Instead it imposes the global CAPM (and checks robustness to Fama–French).

What this buys the authors.

- It sharply reduces noise in mean-return estimates.
- It makes the welfare analysis feasible.
- It ties diversification loss to covariance with a global factor, which is the relevant rewardable risk in the model.

What the cost is. The welfare numbers are only as good as the benchmark asset-pricing model. If the global CAPM is badly wrong, the precise welfare calculations change.

4.3 What identifies the underdiversification decomposition

The empirical decomposition of idiosyncratic risk relies on observed portfolio weights and asset-level return covariances. It does not require the household to rebalance within the sample year.

Interpretation. This is really a measurement exercise: given the observed portfolio at a point in time, what risk would this portfolio imply under the estimated covariance matrix?

4.4 What identifies the nonparticipation cost

There is no experimental variation that tells the authors what nonparticipants would do if they entered. So the welfare cost of nonparticipation is *scenario-based*.

The paper shows:

- under the conventional efficient-entry assumption, the cost is high,
- under a more realistic imputation based on the observed behavior of similar participants, the cost is much lower.

Interpretation. The key identifying assumption here is not an instrument but a behavioral extrapolation: nonparticipants are assumed to resemble cautious and inefficient participants rather than fully efficient investors.

4.5 Why the paper’s claim is persuasive despite the lack of a formal instrument

This is not a modern IV paper. Its credibility comes from:

- unusually rich administrative data,
- transparent accounting decompositions,
- robustness to several alternative benchmark assumptions,
- and the fact that the main qualitative findings line up across many statistics.

In particular, the same households that are less wealthy and less educated:

- participate less,
- take less risk if they participate,
- and appear less efficient if they participate.

That pattern gives the “out might be down” message real bite.

4.6 Main limitations

- Defined-contribution pension wealth is missing.
- Human capital and real estate are not incorporated into the main diversification calculations.
- Expected returns rely on the global CAPM benchmark.
- Mutual fund fees are only roughly incorporated.
- The explanatory power of the regressions linking household characteristics to investment behavior is low.
- The paper is Swedish, so external validity to other countries is suggestive rather than guaranteed.

Bottom line. The results are strongest as a statement about how much we should revise our beliefs once we observe full portfolios, not as a final universal welfare number for every country and period.

4.7 Relation to the literature

The paper sits at the intersection of:

- household portfolio choice,
- limited stock market participation,
- mean-variance diversification,
- and household financial sophistication.

Relative to older work on direct stockholding, its main contribution is to show that mutual funds and cash radically change the interpretation of household mistakes.

5 Tables and figures

5.1 Table 1 and Figure 1: aggregate and cross-sectional asset allocation

	AGGREGATE HOLDINGS (in Billion Dollars)		AGGREGATE ASSET ALLOCATION (from Micro Data)	
	Micro Data (1)	Official Statistics (2)	Wealth Share (3)	Financial Share (4)
Financial assets:				
Bank accounts	46.2	45.3	9.7%	35.1%
Money market funds	7.2	5.3	1.5%	5.5%
Mutual funds	29.4	30.8	6.1%	22.3%
Domestic stocks	27.5	29.3	5.7%	20.9%
International stocks	2.3	NA	.5%	1.8%
Capital insurance	12.1	12.1	2.5%	9.2%
Bonds and derivatives	6.9	8.6	1.4%	5.2%
Total financial assets	131.7	131.3	27.5%	100.0%
Real estate:				
Residential	304.8	284.3	63.7%	
Nonresidential	42.2	60.1	8.8%	
Total real estate	347.0	344.4	72.5%	
Total gross wealth	478.7	475.7	100.0%	
Total debt	147.8	149.1		
Total net wealth	331.0	326.6		
Number of households	4,869,448	4,869,448		
Gross wealth per household	\$98,313	\$97,692		
Net wealth per household	\$67,966	\$67,072		

NOTE.—The table reports aggregate wealth statistics for all resident Swedish households on December 31, 2002. We convert all financial variables into U.S. dollars using the exchange rate at the end of 2002 (1 SEK = \$ 0.1127). In col. 1, we aggregate up the value of the asset holdings observed for each household in our micro data set. Column 2 reports the corresponding official statistic published by Statistics Sweden. We compute in col. 3 the asset allocation of the aggregate portfolio of financial and real estate assets in our data set, and in col. 4 the allocation of the financial portfolio alone.

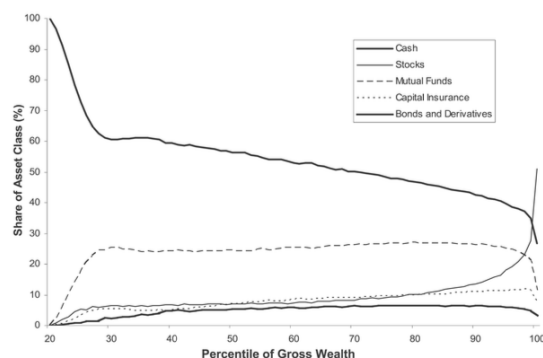


FIG. 1.—Composition of the financial portfolio. The figure represents the cross-sectional distribution of the asset allocation in the financial portfolio owned by Swedish households at the end of 2002. We subdivide households into gross wealth percentiles and report the average asset share within each wealth group. Households in the lowest two deciles are not shown in the figure because their total wealth is poorly measured and they hold almost nothing but cash.

Read. Table 1 shows the aggregate Swedish household balance sheet in 2002. Figure 1 shows how financial wealth composition varies by wealth percentile.

- Real estate is most of household gross wealth.
- Within financial wealth, cash is large, and mutual funds are as important as direct stocks in the aggregate.
- Figure 1 shows that mutual funds dominate risky wealth for most households, while direct stocks dominate only in the top tail.

	ALL HOUSEHOLDS			PARTICIPANTS			NONPARTICIPANTS		
	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation
Portfolio characteristics									
Complete portfolio (\$)	...	...	...	34,595	12,338	228,986	...	...	...
Risky portfolio (\$)	...	...	...	18,694	4,295	174,997	...	...	...
Stock portfolio (\$)	...	...	...	8,545	200	133,475	...	...	...
Financial characteristics									
Disposable income (\$ per year)	26,135	20,985	28,861	30,948	26,229	34,647	18,092	15,195	16,747
Private pension premia/income (%)	1.26	.00	6.15	1.70	.19	7.41	.52	.00	2.93
Financial wealth (\$)	28,323	7,069	193,692	40,504	14,389	243,507	7,970	1,270	21,511
Log financial wealth	8.83	8.86	1.72	9.54	9.57	1.46	7.64	7.15	1.45
Real estate wealth (\$)	75,357	26,710	208,228	102,983	59,869	253,653	29,194	0	73,507
Log real estate wealth	6.60	10.19	5.59	8.18	11.00	5.19	3.97	.00	5.23
Total liability (\$)	32,156	9,222	123,177	40,625	16,608	98,274	18,004	2,876	155,139
Log total liability	6.74	9.13	4.75	7.33	9.72	4.71	5.75	7.96	4.67
Retirement dummy	.26	.00	.44	.23	.00	.42	.32	.00	.47
Unemployment dummy	.08	.00	.28	.07	.00	.25	.11	.00	.31
Entrepreneur dummy	.03	.00	.17	.04	.00	.19	.02	.00	.13
Student dummy	.04	.00	.20	.04	.00	.19	.05	.00	.22
Demographic characteristics									
Age	52.19	51.00	18.50	51.16	50.00	17.24	53.92	52.00	20.33
Household size	1.94	1.00	1.24	2.17	2.00	1.29	1.55	1.00	1.05
High school dummy	.64	1.00	.48	.71	1.00	.45	.52	1.00	.50
Post-high school dummy	.25	.00	.43	.31	.00	.46	.15	.00	.36
Dummy for unavailable education data	.17	.00	.37	.13	.00	.33	.24	.00	.43
Immigration dummy	.15	.00	.35	.11	.00	.31	.21	.00	.41

Note.—The table reports summary statistics of the main financial and demographic characteristics of Swedish households at the end of 2002. We convert all financial variables into U.S. dollars using the exchange rate at the end of 2002 (1 SEK = \$ 0.1127). The computations are based on the random sample considered throughout the empirical analysis. Missing bank account balances are imputed using the methodology outlined in the online Appendix. All logarithms are computed in the natural base and truncated at zero.

Why it matters. This is the first warning against overinterpreting direct-stock concentration. If most households hold diversified mutual funds, then the stock-only picture is misleading.

5.2 Table 2: who participates and how large are portfolios?

Read. Table 2 summarizes financial and demographic characteristics of all households, participants, and nonparticipants.

Main numbers to remember:

- participation rate about 62%,
- median risky portfolio among participants about \$4,295,
- participants are richer, more educated, and have higher income.

Why it matters. Participation is strongly selected. This already hints that the cost of nonparticipation cannot be computed under the assumption that nonparticipants would look like the average efficient participant.

5.3 Table 3 and Figure 2: how risky and how inefficient are risky portfolios?

Read. Table 3 decomposes total risky-portfolio volatility into systematic and idiosyncratic components. Figure 2 places household portfolios in mean–standard deviation space relative to the world benchmark and the Swedish index.

Key results:

- Median risky-portfolio volatility: 19.5%.
- Median systematic risk: 13.0%.
- Median idiosyncratic risk: 14.4%.
- Median idiosyncratic variance share: 54.9%.
- Median beta with the hedged world index: about 0.89.

Interpretation.

- The typical risky portfolio is not on the efficient frontier.

	CROSS-SECTIONAL DISTRIBUTION									
	Mean	1st Percentile	5th Percentile	10th Percentile	25th Percentile	50th Percentile	75th Percentile	90th Percentile	95th Percentile	99th Percentile
A. Total Volatility										
Total risk $\sigma_{i,t}$ (%)	22.2	4.7	7.5	11.3	16.0	19.5	24.0	36.4	49.1	64.5
Systematic risk $[\beta_i/\sigma_p]$ (%)	12.8	6	4.6	7.8	11.2	13.0	14.4	17.9	22.3	24.4
Idiosyncratic risk $\sigma_{i,t}$ (%)	17.6	4.6	5.9	8.2	11.4	14.4	18.9	31.9	43.5	62.9
Idiosyncratic share $(\sigma_{i,t}/\sigma_p)^2$ (%)	60.8	97.9	61.3	52.3	39.9	54.9	62.8	77.5	78.3	83.2
Beta coefficient β_i	.87	.04	.32	.53	.76	.89	.98	1.16	1.52	1.65
B. Idiosyncratic Volatility										
Idiosyncratic risk $\sigma_{i,t}$ (%)	17.6	4.6	6.0	8.2	11.3	14.7	18.9	31.2	42.6	57.6
Asset volatility $\sigma_{i,t}$ (%)	21.1	5.4	8.3	8.6	14.6	18.5	26.6	35.5	48.8	70.2
Concentration $C_{i,t}$ (%)	.37	.82	.32	.91	.35	.46	.37	.72	.68	.67
Asset correlation $\rho_{i,t}$	.25	.04	.12	.14	.35	.29	.21	.14	.09	.13
Stock share $D_{i,t}$	.28	.01	.04	.01	.09	.16	.32	.90	.90	.82

Notes.—The first column of each panel reports the sample mean of portfolio characteristics among participating households. In the text set of columns, households are sorted by their level of total risk (panel A) or idiosyncratic risk (panel B), and the mean of 500 households around the corresponding percentile is reported.

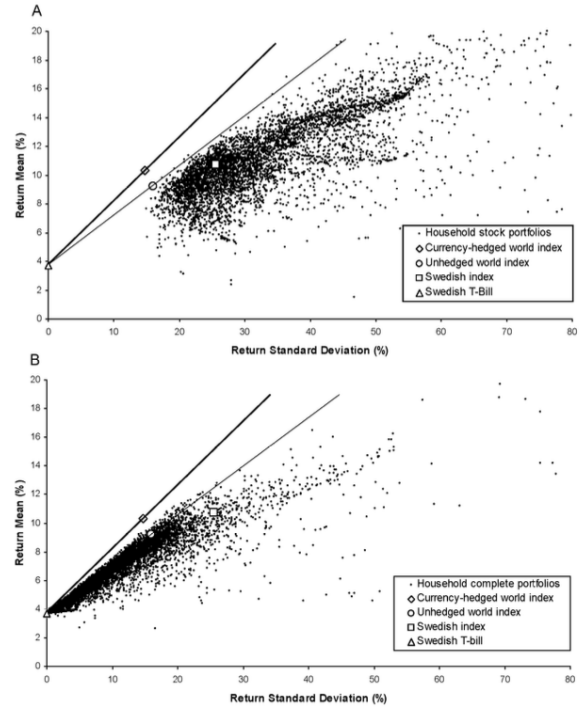


FIG. 2.—Scatter plots of household portfolios. A, Stock portfolios. The scatter plot illustrates the mean and standard deviation of household stock portfolio returns. B, Complete portfolios. The scatter plot illustrates the mean and volatility of household complete portfolios. The mean returns are inferred from the global CAPM, in which the currency-hedged world index (empty diamond) is mean-variant efficient. The graphs are based on a random sample of 10,000 households at the end of 2002.

- But Figure 2B shows that complete portfolios often lie reasonably close to the efficient line because households combine moderate-risk risky portfolios with a lot of cash.

Why it matters. This is the core distinction between a bad risky portfolio and a badly chosen *complete* portfolio. The latter is what matters for welfare.

5.4 Table 4: diversification losses

BENCHMARK	CROSS-SECTIONAL DISTRIBUTION							
	Mean	25th Percentile	50th Percentile	75th Percentile	90th Percentile	95th Percentile	99th Percentile	
A. Complete Portfolio Relative Sharpe Ratio Loss								
Currency-hedged world index	.38	.39	.35	.42	.35	.69	.89	
Unhedged world index	.19	.07	.14	.24	.41	.60	.85	
Sweden index	-.02	-.16	-.08	.04	.26	.49	.82	
B. Complete Portfolio Return Loss (%)								
Currency-hedged world index	1.68	.54	1.17	2.06	3.40	5.04	9.86	
Unhedged world index	.66	.09	.30	.71	1.38	2.65	5.84	
Sweden index	-.01	-.36	-.11	.05	.61	1.17	3.28	

C. Risky Portfolio Return Loss (%)								
Currency-hedged world index	4.14	2.01	2.92	4.21	8.51	12.16	17.91	
Unhedged world index	1.75	.40	.87	1.73	4.61	7.07	11.25	
Sweden index	.16	-.82	-.39	.22	1.93	3.20	7.52	
D. Return Loss in Dollars								
Currency-hedged world index	740	36	131	433	1,190	2,204	7,566	
Unhedged world index	321	6	33	133	426	851	3,254	
Sweden index	43	-55	-8	4	78	218	1,129	
E. Return Loss as a Fraction of Disposable Income (%)								
Currency-hedged world index	2.13	.14	.51	1.64	4.48	7.84	23.99	
Unhedged world index	.86	.02	.13	.52	1.62	3.13	11.42	
Sweden index	.02	-.20	-.03	.02	.32	.89	4.48	

Notes.—The table reports the cross-sectional distribution of household diversification losses with respect to three benchmark indexes: the currency-hedged world index, the unhedged world index, and the MSCI Sweden index. Losses are computed in yearly units according to five different measures: (1) the relative Sharpe ratio loss on the complete portfolio; (2) the return loss on the complete portfolio; (3) the return loss on the risky portfolio; (4) the return loss in dollars; and (5) the return loss as a fraction of disposable income. The dollar loss is obtained by multiplying the return loss on the risky (or complete) portfolio by the dollar value of the risky (complete) portfolio. The return loss as a fraction of disposable income is obtained by dividing the dollar loss by the three-year average of household disposable income. The mean and standard deviation of asset returns are computed using the global CAPM.

Read. Table 4 reports relative Sharpe-ratio losses and return losses using three benchmarks: the currency-hedged world index, the unhedged world index, and the Swedish domestic index.

Main numbers:

- Median relative Sharpe-ratio loss: 35% relative to the hedged world index, 14% relative to the unhedged world index.

- Median complete-portfolio return loss: 1.17% relative to the hedged world index, 0.30% relative to the unhedged world index.
- Median dollar loss: about \$131 (hedged benchmark) and \$33 (unhedged benchmark).
- 95th percentile complete-portfolio return loss: about 5.04%.

Interpretation.

- The median household loses something, but not a huge amount.
- The right tail is where the serious mistakes live.
- A majority of Swedish households even outperform the Sharpe ratio of their domestic stock index because they diversify internationally through funds.

Why it matters. The paper’s headline conclusion comes from this table: typical households are not doing nearly as badly as the earlier direct-stock literature suggested.

5.5 Table 5 and Figure 3: sophistication, aggressiveness, and losses

TABLE 5
CONTRIBUTORS TO COMPLETE RETURN LOSS

	RETURN LOSS $\ln(R_{t-\text{complete}})$			RISKY SHARE $\ln(\alpha)$			RISKY PORTFOLIO BETA $\ln[\beta]$			DIVERSIFICATION LOSS $\ln[\text{RSRL}/(1 - \text{RSRL})]$		
	Estimate	tStatistic	Change	Estimate	tStatistic	Change	Estimate	tStatistic	Change	Estimate	tStatistic	Change
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Financial characteristics:												
Disposable income	.007	3.49	2.1%	-.007	-3.83	-2.1%	.009	9.12	2.7%	.005	4.79	1.5%
Private pension premium income	.248	3.36	1.8%	.351	4.92	2.6%	-.016	-.44	-.1%	-.087	-2.17	-.6%
Log financial wealth	.090	20.20	14.1%	.137	31.90	22.3%	-.016	-6.93	-2.2%	-.032	-13.00	-4.5%
Log real estate wealth	.008	7.97	5.1%	.005	3.21	3.2%	.003	6.63	2.1%	.000	-.75	-.3%
Log total liability	.012	9.40	7.0%	.004	3.21	2.3%	.010	15.00	5.6%	-.002	-9.29	-.9%
Retirement dummy	-.043	-1.95	-4.2%	-.023	-1.10	-2.3%	-.050	-4.53	-4.9%	.031	2.53	3.1%
Unemployment dummy	-.086	-4.01	-8.2%	-.105	-5.06	-9.9%	-.001	-.11	-.1%	.020	1.72	2.0%
Entrepreneur dummy	-.115	-4.01	-10.8%	-.261	-9.40	-22.9%	.097	6.75	10.2%	.049	3.11	5.0%
Student dummy	.020	.65	2.0%	.069	2.32	7.1%	-.053	-3.44	-5.2%	.004	-.24	-.4%

Demographic characteristics:

Age	-.001	-1.97	-1.9%	-.001	-2.01	-1.9%	-.002	-7.56	-3.7%	.002	6.90	3.8%
Household size	-.143	-29.30	-16.9%	-.086	-18.20	-10.5%	-.010	-4.13	-1.3%	-.047	-17.60	-5.9%
High school dummy	.111	6.87	10.5%	.107	6.87	10.2%	.057	7.06	5.6%	-.053	-6.07	-5.5%
Post-high school dummy	.173	13.20	18.9%	.124	9.81	13.2%	.042	6.47	4.3%	.006	.89	.6%
Dummy for unavailable education data	.112	4.62	11.9%	.087	3.68	9.1%	-.037	-3.04	-3.7%	.063	4.74	6.5%
Immigration dummy	.043	2.53	4.4%	-.112	-6.76	-10.6%	.045	5.26	4.6%	.110	11.80	11.6%
Intercept	-1.093	-20.00	...	-2.751	-52.20	...	-.108	-3.95	...	-.127	-4.27	...
Adjusted R ²	.03			.04			.05					

NOTE.—This table reports the OLS regression of the log complete return loss and in three components onto observable household characteristics. Losses are computed relative to the hedged world index. For each regression, we report the linear coefficient, standard deviation, and marginal effect of each predicting variable. The marginal effect is assessed by computing the impact on the dependent variable (in levels) of increasing a continuous regressor by one standard deviation, or of setting a dummy variable equal to one. Since

$$\ln(R_{t-\text{complete}}) = \ln(\beta C) + \ln(\alpha) + \ln[\beta] + \ln[\text{RSRL}/(1 - \text{RSRL})],$$

the coefficient estimate for the complete loss in each row is the sum of the other three coefficient estimates (with the exception of the intercepts in the last row). Financial variables are expressed in Swedish kronor, and disposable income is rescaled as a multiple of 100,000 SEK. All logarithms are computed in the natural base and winsorized at zero.

Read. Table 5 decomposes complete return loss into risky share, beta, and diversification inefficiency. Figure 3 gives the intuition graphically.

Main message:

- Wealth, education, and related proxies for sophistication predict lower Sharpe-ratio loss (better risky portfolios),
- but they also predict much higher risky shares,
- so complete return loss rises with sophistication.

Figure 3 intuition. A financially sophisticated household holds a risky portfolio closer to the efficient frontier, but it chooses much more overall risk, so its vertical distance from the efficient line at the chosen risk level can still be larger than that of an unsophisticated household that barely takes any risk.

Why it matters. This is one of the paper’s most subtle points. “Better investors” are not automatically those with smaller complete-portfolio losses.

5.6 Table 6: who stays out, and what would they do if they came in?

Read. Table 6 reports the probit for participation and OLS regressions for the risky share and Sharpe ratio among participants.

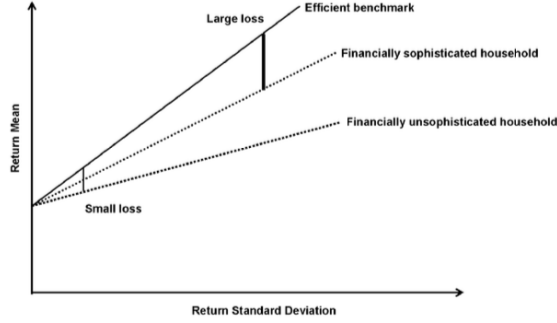


FIG. 3.—Impact of financial sophistication on the complete return loss. Rich and educated households select portfolios with a high Sharpe ratio but also a high risky share, resulting in a high complete return loss. Conversely, unsophisticated households allocate a small fraction of their financial wealth to an inefficient risk portfolio and overall incur low complete portfolio return losses.

TABLE 6
PARTICIPATION AND RISKY ASSET SHARE

	PARTICIPATION: Probit Regression			COMPLETE PORTFOLIO RISK: α/α_1			SHARPE RATIO: $\bar{S}_i$		
	Estimate (1)	t-Statistic (2)	Change (3)	Estimate (4)	t-Statistic (5)	Change (6)	Estimate (7)	t-Statistic (8)	Change (9)
Financial characteristics:									
Disposable income	.106	15.30	8.8%	.096	7.38	3.1%	-.001	-5.24	-.6%
Private pension premia/income	3.053	14.00	6.1%	.533	1.08	.4%	.009	2.24	.2%
Financial wealth (in log)	.491	115.00	22.5%	.566	18.90	9.0%	.003	12.50	1.6%
Real estate wealth (in log)	.021	22.80	4.6%	.026	3.78	1.7%	.000	.75	.1%
Total liability (in log)	.017	13.50	3.3%	.117	13.50	7.1%	.000	1.92	.3%
Retired dummy	.137	6.04	4.6%	.247	-2.33	-3.8%	-.003	-2.49	-1.0%
Unemployment dummy	-.065	-3.61	-2.3%	-.019	-.14	-.2%	-.002	-1.92	-.8%
Entrepreneur dummy	-.030	-.94	-1.0%	-.011	-.06	-.1%	-.005	-3.42	-1.8%
Student dummy	.028	1.08	1.0%	.079	.38	.9%	.000	-.08	.0%
Demographic characteristics:									
Age	-.014	-26.30	-9.8%	-.025	-6.38	-4.6%	.000	-6.78	-1.2%
Household size	-.009	-1.62	-.4%	-1.112	-34.00	-15.6%	.005	18.70	2.2%
High school dummy	.195	13.50	7.1%	.718	6.62	7.8%	.005	6.21	1.8%
Post-high school dummy	.130	9.12	4.3%	1.119	12.70	12.1%	-.001	-1.22	-.3%
Dummy for unavailable education data	-.066	-3.04	-2.3%	.731	4.47	7.9%	-.006	-4.47	-2.0%
Immigration dummy	-.384	-26.30	-14.4%	.873	7.59	9.5%	-.011	-12.60	-4.0%
Intercept	-4.852	-98.80	...	3.982	10.90	...	.244	83.10	...
Adjusted R^2				.04			.03		

Notes.—Columns 1-3 report a probit regression of the participation decision on financial and demographic household characteristics. Participation is defined as the ownership of at least one risky asset. Columns 4-6 report an OLS regression of the participants' risky share on the same set of observable characteristics. The risky share is defined as the weight of risky assets in the complete portfolio. For each regression, we report the linear coefficient, standard deviation, and marginal effect of each predicting variable. The marginal effect is assessed by computing the impact on the dependent variable of increasing a continuous regressor by one standard deviation, or of setting a dummy variable equal to one. The reported change is the absolute increase in the probability of participation for the probit and the percentage change in the dependent variable for the other two regressions. Financial variables are expressed in Swedish kronor, and disposable income is recoded as a multiple of 100,000 SEK. All regressions are computed in the natural base and vintennized at zero.

Main participation patterns:

- higher income, wealth, education, liabilities, and pension contributions predict participation,
- age, larger household size, unemployment, and immigration reduce participation,
- financial wealth is the strongest predictor.

The table is then used to impute likely risky-share and Sharpe-ratio choices for nonparticipants.

Why it matters. This is the empirical bridge from “who is out” to “would they be down if they entered?”

5.7 Section V calculations: welfare cost of nonparticipation

Read. The paper compares two scenarios.

1. If nonparticipants entered and invested efficiently, the return cost of nonparticipation is about 4.3%.
2. If they entered and invested like similar cautious and inefficient participants, the cost falls to about 2.3%.

Why it matters. This is where the title of the paper comes alive. Households that are currently out may stay out partly because entering would not make them efficient investors.

6 Short summary

- The paper uses unusually rich Swedish administrative data to measure household diversification at the level of the entire risky portfolio.
- Once mutual funds and cash are included, the median household is not disastrously underdiversified. The median complete-portfolio return loss is only about 1.17% of financial wealth, or roughly \$131 per year relative to the currency-hedged world index.
- The minority in the right tail is different: some households incur very large losses from concentrated risky portfolios.

- Richer and more educated households invest more efficiently in the sense of lower Sharpe-ratio losses, but they also invest more aggressively, so their complete-portfolio return losses can be higher.
- Nonparticipants are not just people missing out on the equity premium. They also look like households who would likely invest cautiously and inefficiently if they entered. This cuts the estimated cost of nonparticipation from about 4.3% to about 2.3%.
- The big lesson is that household mistakes must be measured using complete portfolios and realistic counterfactual behavior, not idealized benchmark assumptions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes three main points.

First, measuring household diversification using only directly held stocks is misleading. Once mutual funds and cash are observed, most Swedish households appear reasonably well diversified, even though a minority is badly underdiversified.

Second, underdiversification and nonparticipation are linked. The households most likely to stay out of risky markets are also the households most likely to invest inefficiently if they entered.

Third, the welfare cost of household mistakes is therefore more nuanced than the standard participation puzzle suggests: the median underdiversification loss is small, the right-tail loss is large, and the cost of nonparticipation is substantially smaller once realistic investment performance is imputed.

7.2 Economic intuition

The economic intuition is simple but powerful.

A household can make two distinct mistakes:

1. it can choose a bad risky portfolio,
2. or it can choose the wrong amount of risk.

The paper shows that many unsophisticated households make the first mistake but partly offset it by avoiding the second: they hold inefficient risky portfolios, but they hold very little of them. By contrast, more sophisticated households choose better risky portfolios but take much more risk. This is why complete-portfolio return loss is not monotone in sophistication.

For nonparticipants, the same logic matters. The foregone equity premium is not the right welfare concept unless one assumes that these households would invest efficiently if they entered. If they would in fact enter with cautious, poorly diversified portfolios, then staying out is less costly than the standard model says.

7.3 Takeaway

The paper's deepest takeaway is methodological: to evaluate household finance mistakes, we need the *right object* and the *right counterfactual*.

The right object is the whole portfolio, not just directly held stocks. The right counterfactual is not “what if everyone invested like an efficient representative agent?” but “what would this household likely do if it changed its behavior?”

Using the right object and the right counterfactual turns a dramatic claim — “households are making huge mistakes” — into a more precise one: *most households do okay, some do very badly, and the households that stay out might well do badly if they came in.*